

Bug Hunt 4: Polynomial

What the program does

`poly.erl` represents and manipulates polynomials with integer coefficients. A polynomial is a list of `{Exponent, Coefficient}` pairs, with these invariants:

- No two terms share the same exponent.
- No term has coefficient 0 (those are dropped).
- The order of the list is not significant — `[{3,2}, {1,-5}, {0,7}]` and `[{0,7}, {3,2}, {1,-5}]` both represent the same polynomial, $2x^3 - 5x + 7$.

The module exports:

- `evaluate(P, X)` — compute $P(X)$, the value of P at integer X .
- `derivative(P)` — return the symbolic derivative $P'(X)$.
- `add(P, Q)` — return $P+Q$, merging terms with equal exponents and dropping any term whose coefficients sum to 0.
- `scale(P, K)` — multiply every coefficient by K .

`main.erl` exercises these operations on $p(x) = 2x^3 - 5x + 7$ and $q(x) = x^2 + 3$. It evaluates p at several points, takes the derivative, adds polynomials, scales, and checks the algebraic identity $p + (-p) = 0$.

The catch

This program contains **exactly 3 bugs** that prevent it from producing the expected output. The bugs are all *semantic* — the program compiles without errors and runs to completion without crashing.

You may not run the program until you have submitted your answer. Trace by reading.

Expected output

```
p(x) = [{3,2},{1,-5},{0,7}]
q(x) = [{2,1},{0,3}]

p(0) = 7
p(1) = 4
p(2) = 13
p(-1) = 10
```

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```
p'(x) = [{2,6},{0,-5}]
p'(2) = 19

p + q = [{2,1},{0,10},{1,-5},{3,2}]
3p    = [{3,6},{1,-15},{0,21}]

-p     = [{3,-2},{1,5},{0,-7}]
p + (-p) = []
```

Note that $p + q$ is not in any particular order; the term-list order depends on how `add` walks the inputs. Don't treat the order as a bug.

Your write-up

For each bug, record:

1. **Location** — file name and line number(s).
2. **Diagnosis** — in 1–2 sentences, what is wrong.
3. **Fix** — the corrected line(s).
4. **Symptom** — which line(s) of the expected output differ from what the buggy program actually prints, and why this specific bug causes that specific symptom.